Ink & Steel: The librarian's Tale GDD + LDD + TDD

By Daniel Isidoro Ramírez Villalta

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1. Game Design Document

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1.1. High Concept

“Ink & Steel: The librarian’s tale” is a comedic fantasy turn-based JRPG. The story follows a librarian who accidentally ends up sharing a body with a mercenary and must escape the book he has been trapped in. Gameplay focuses on a dual-control mechanic: following the mercenary’s strategic orders while the librarian is in command, or letting the mercenary take full control of the body to unleash his combat expertise.

1.2. Mechanics & Dynamics

Common Actions (World Exploration):

- Walk: Basic character movement through the environment.
- Run: Increased movement speed while holding the sprint button.
- Interact: Contextual action to engage with NPCs, objects, or environmental elements.
- Vault / Step Over: Contextual interaction to move over low-height obstacles (e.g., debris, small furniture).

The Librarian’s Actions (Combat Phase):

- Attack: Basic offensive action that builds the "Synergy" meter.
- Defend/Evade: Reactive actions triggered via QTE (Quick Time Event).
- Successful Block: Reduces incoming damage and significantly grants a large Synergy boost.
- Successful Evade: Nullifies all damage and grants a small Synergy boost.
- Listen: Follow the Mercenary’s mental advice to gain attribute buffs for one or more turns.

The Mercenary’s Actions (Dominance Phase):

- Dominate: Once the Synergy meter is full, the Mercenary takes control, significantly increasing all character stats.
- Combat Skills: Access to powerful high-damage abilities that consume the accumulated Synergy meter.

Core Interaction Flow:

- Combat always begins with the Librarian in control. Players must focus on defensive timing (blocks/evades) and basic attacks to fill the Synergy meter. Upon reaching 100%, the Mercenary can "Dominate" the body, shifting the gameplay from a survival/management style to a high-power offensive style.

1.2.1. Gameplay Example: Combat Encounter

Encounter: The Librarian vs. a Goblin.

Librarian Phase (Survival & Synergy Building):

- **Active Offense:** Combat begins with the Librarian attacking through a combo system. Success increases the Synergy meter based on combo length. Visuals reflect the character's clumsiness: attacks look accidental, as if hitting by pure luck. A

critical timing failure may cause the Librarian to trip or fall, immediately breaking the combo.
- **Reactive Defense:** When the Goblin attacks, the player chooses between Evade or Defend. After a successful initial block/evade, the Mercenary's voice guides the Librarian through the rest of the enemy's combo, helping to sustain the defense. Movements remain awkward and seemingly "lucky" in execution.

Transition & Mercenary Phase (Dominance):

- **The "Dominate" Command:** Once the Synergy meter reaches 100%, the player triggers the **Dominate** state. The Mercenary takes full control of the body.
- **Professional Mastery:** All stats increase significantly. Attack combos deal more damage, and defensive timings become more forgiving. The character's animations shift from clumsy to professional and precise. In this state, combos are no longer interrupted by timing errors, reflecting the Mercenary's combat expertise.

1.3. Target Audience

Based on Bartle's Taxonomy of Player Types, our primary target audience consists of:

- **Explorers:** Players seeking original narratives and comedic storytelling.
- **Achievers:** Players who enjoy mastering complex combat mechanics and optimizing character builds. The game balances initial "comedic frustration" with a satisfying sense of late-game power progression.

1.4. Core Loop

Action: The player enters combat, initially at a disadvantage, relying on weak and basic survival movements.
Reward: Successfully performing actions fills the Synergy meter. The Mercenary provides positive feedback as the meter grows. Once full, the player triggers the "Dominance" state, executing lethal moves to finish the encounter.
Progression: Victory grants experience points and items to boost stats and unlock new abilities.

- **The Librarian** becomes more competent and less clumsy over time.
- **The Mercenary** unlocks devastating new skills. The journey continues through a chapter-based narrative structure.

1.5. Narrative & World-Building

1.5.1. World Narrative

The game is set in a classic medieval fantasy world—kings, castles, and magic—but with a meta-fictional twist. The protagonist, a Librarian from a mystical archive, accidentally traps himself inside one of the books he guards, sharing a body with a solitary Mercenary. Together, they must navigate the story's clichés to reach the book's conclusion and escape. The Librarian must balance following the "written script" to avoid fatal consequences while surviving the Mercenary's impulsive nature.

1.5.2. Character Arcs

- **The Librarian & The Mercenary:** They start as polar opposites—intellectual and cowardly vs. brutal and brave. As the story progresses, they forge a bond, learning from each other. The Librarian finds his courage to fight for what is right (even if it changes the script), while the Mercenary learns the value of strategy and teamwork.

1.5.2. Key NPCs

The Narrator: A guiding voice only the protagonists can hear. He becomes frustrated and may argue with the players if they deviate from the written story.
The White Knight: The "official" hero of the book—a walking cliché who finds himself overwhelmed when the protagonists' interference changes his predestined path.
The Writer (Antagonist): A former Librarian trapped in the book long ago. He "edits" the world in real-time to suit his dark purposes.
Blot (Borrón): A sentient ink stain that acts as a mysterious merchant, offering unique items and gear that "don't belong" in the original story.

1.6. Visual Style

Art Direction: 32-bit Pixel Art with a modern HD-2D approach for environments and backgrounds. Character sprites feature a more detailed "Chibi" style during combat.
Diegetic Interface: The screen features a subtle, semi-transparent frame mimicking book pages. Transitions between rooms or scenes use a "page-turn" effect (forward or backward) to reinforce the feeling of being trapped inside a book.
Color Palette as Feedback:

- **Librarian Mode:** Blue and Lilac palette for menus and UI when the Librarian is in control or speaking.
- **Mercenary Mode:** Red and Orange palette for menus and UI during "Dominance" or when the Mercenary speaks.

Combat Indicators:

- **Health Bar:** Dynamic green bar that turns grey as health depletes.
- **Synergy Meter:** Yellow.
- **Visual Cues:** White flashes for QTE timing (combos/defense) and Red flashes for unblockable attacks (indicating "Evade Only").

1.6.1. Visual Examples

Narrative Transitions: Upon completing a dungeon and moving to a new area, a "page-flip" animation plays, signaling progress in the story. Returning to a previous area triggers the reverse animation, emphasizing the book-world theme.
UI Shifting: While controlling the Librarian, the selection menu is Blue/Lilac. Upon using the "Dominate" command, the character's idle stance changes to a professional combat posture, and the entire UI shifts to Red/Orange, providing instant visual feedback of the power-up.
Combat Clarity: During an attack, a white glint indicates the exact frame to continue a combo. If an enemy prepares a powerful move, a red glint warns the player that blocking is impossible, forcing a tactical shift to evasion.

1.7. Audio Identity

Musical Style: Orchestral soundtrack with a Medieval Folk influence.
Dynamic Music System: The score adapts to the gameplay state.

- **Librarian Phase:** Woodwinds and strings dominate in a lighthearted, comedic tone.
- **Mercenary Phase (Dominance):** The same melody transitions seamlessly into brass and epic percussion, increasing the tempo to reflect a more frenetic combat pace.

SFX & Ambience: 32-bit synthesized sound effects.

- **Audio Feedback:** A "bright" chime for successful QTE timings and a "muted" thud for failures.
- **Environmental Audio:** Contextual layers including wind, crackling fire, distant war cries, and wildlife to deepen immersion.

1.8. Screen Flow & UI/UX

Menu Navigation: The UI is integrated into the "Book" theme. Options like "New Game" or "Load" appear as chapters in a table of contents.
Transitions: Loading a game involves a rapid page-flip animation and a zoom-in effect, symbolizing the player "entering" the story.
UI Categories:

- **Diegetic:** The **Save Book**, an in-game item the Librarian carries to record enemy weaknesses and save progress.
- **Non-Diegetic:** Health and Synergy bars providing essential combat data.
- **Spatial:** QTE indicators appearing directly over the characters/weapons during action sequences.
- **Meta:** The screen edges subtly mimic an open book, maintaining the "trapped in a story" aesthetic at all times.

1.8.1. UX Examples

Visual Mode Shift: When the Synergy meter is full, the "Dominate" command glows. Upon activation, the Librarian’s blue menu "burns away" like paper to reveal the Mercenary’s aggressive red/orange interface.
Bestiary: Accessing the enemy list opens a physical book index; selecting an entry flips the pages directly to that enemy’s specific profile.

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2.1. Level Metrics

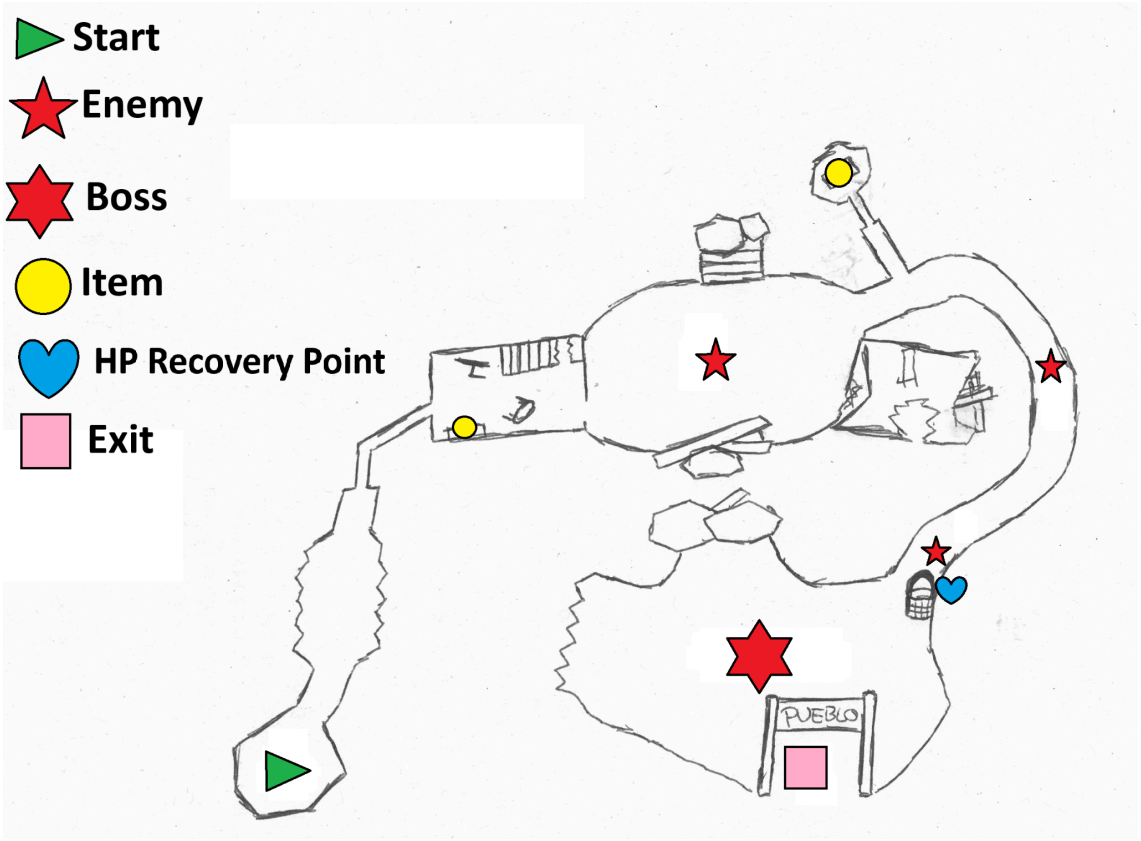
Measurement Units: Grid-based system using 32x32 pixel tiles.
Character Scale: The protagonist stands at 1.5 tiles in height.
Environmental Interaction: While there is no dedicated jump button, the character can vault or step over obstacles up to 1 tile high (e.g., broken furniture, chairs).
Movement Speeds:

- **Walk:** 3 tiles per second.
- **Run:** 5 tiles per second.

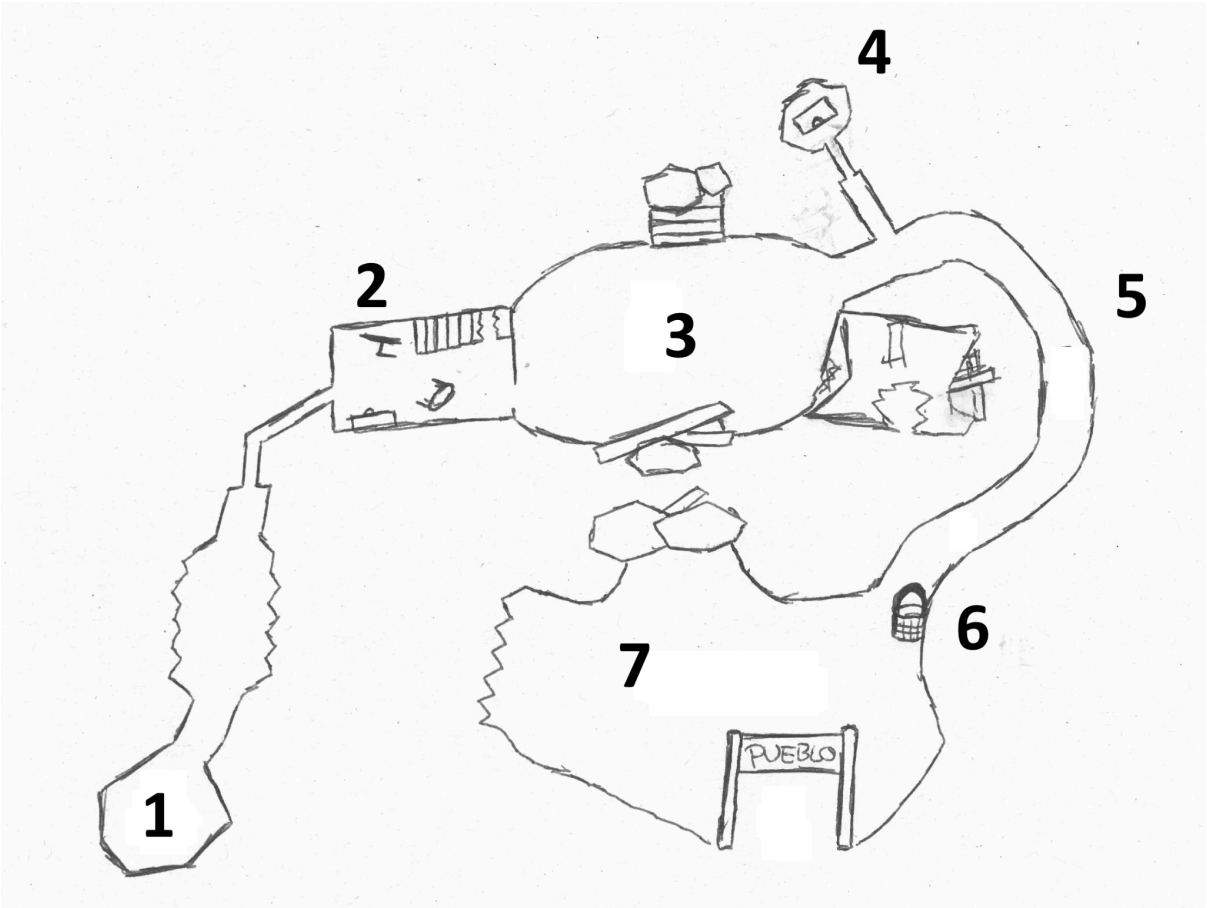
Level Pacing: The initial stage focuses on a deliberate, linear path to allow players to familiarize themselves with the controls and movement mechanics.

2.2. Level Layout & Flow

Structure: The map follows a linear progression with minor branching paths for exploration.
Pacing Strategy: The level begins in a Safe Zone, gradually transitioning into more open but controlled areas.
Intensity Curve: The player encounters "tutorial-style" combat encounters that escalate in difficulty, leading toward a final climax (Boss encounter).



2.3. Pacing & Level Progression (Beat Sheet)



1. **Narrative Intro:** The Librarian wakes up in an unfamiliar body. Initial dialogue with the Mercenary (internal voice) establishes the conflict without knowing if that voice is real. The player learns basic movement and "tight-space" navigation while moving through a battlefield.
2. **Environmental Interaction:** Introduction to looting and searching. Tutorial prompt encourages the player to search cabinets and furniture for hidden items.
3. **Scripted Combat (Tutorial):** A "can't-lose" encounter to teach basic turn-based commands. The Librarian realizes the Mercenary is still conscious within the same body.
4. **Optional Exploration:** Introduction to "hidden" or optional paths/items to reward curious players.
5. **Practice Encounters:** A series of guided skirmishes where the Mercenary provides tactical advice to solidify combat mechanics.
6. **Boss Climax:** A final challenge where the player must apply everything learned. This encounter triggers the first use of the "Dominate" command, showcasing the Mercenary’s true power.

2.4. Asset List

Graphics & Sprites:

- **Characters:** Mercenary (Librarian), Attacking Soldier, Defending Soldier, Captain.
- **Environment:** Ruined walls/houses, war stakes, debris (broken tables/chairs/ladders), trees, and rocks.
- **Interactables:** Chests, cabinets, doors.

Audio & SFX:

- **Dynamic Music:**
    - *Ambient:* War-themed orchestral folk.
    - *Battle:* War theme with heavy percussion.
    - *Boss:* Orchestral Rock.
    - *Dominate State:* Orchestral Rock + Epic Brass sections.
- **SFX:** Footsteps (dirt), cabinet/door opening, sword-on-shield impacts, fire crackling, and the signature "Page-Flip" sound for scene transitions.

3. Technical Design Document

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3.1. Script Architecture

The project is structured into modular scripts to ensure scalability:

- **PlayerController:** Core logic, inventory, and pause management.
- **MovementSystem:** Handles grid-based translation and speed states.
- **InteractionManager:** Contextual triggers and NPC dialogue systems.
- **EnemyAI:** Patrol, pursuit, and combat engagement logic.
- **CombatEngine:** Turn-based mechanics, QTE processing, and stat calculations.
- **AudioManager:** Dynamic music transitions and environmental SFX.

3.2. Systems Logic (Pseudocode)

Character & Global Logic:

```
Inventory and Pause management
IF inventory_button == TRUE AND is_inventory_open == FALSE:
   open_inventory()
ELSE:
   close_inventory()

IF pause_button == TRUE AND is_game_paused == FALSE:
   pause_game()
ELSE:
   resume_game()

IF player_vitality < 0:
   trigger_game_over()
```

Movement System:

```
Walking
IF direction_arrow == TRUE:
   x_axis += 3 OR x_axis -= 3 // Also for y_axis

Running
IF direction_arrow == TRUE AND run_button == TRUE:
   x_axis += 5 OR x_axis -= 5 // Also for y_axis
```

Interaction & Environment:

```
Vaulting over small obstacles
IF interact_pressed AND object_height <= 32px:
   is_vaulting = TRUE

NPC Dialogue
IF interact_pressed AND npc_detected == TRUE:
   start_conversation()
```

Enemy Behavior (Overworld):

```
IF distance_to_player <= 12:
   pursue_player()

IF contact_with_player == TRUE:
   start_combat_engine()

IF combat_victory == TRUE:
   delete_enemy_instance()
```

Combat Engine (Turn-Based Logic):

```
IF start_combat:
   WHILE player_hp > 0 AND enemy_hp > 0:
      // PLAYER TURN
      IF player_action == ATTACK:
         IF qte_success:
            damage = (player_atk * 1.5) - enemy_def
         ELSE:
            damage = player_atk - enemy_def

         enemy_hp -= MAX(0, damage) // Prevents negative damage

      IF player_action == INVENTORY:
         open_combat_inventory()

      IF player_action == FLEE:
         end_combat()

      // ENEMY TURN
      IF enemy_action == ATTACK:
         IF player_evades:
            IF qte_success: player_hp -= 0
            IF qte_fail: player_hp -= MAX(0, (enemy_atk * 1.5) - player_def)

         IF player_blocks:
            IF qte_success: player_hp -= MAX(0, (enemy_atk * 0.5) - player_def)
            IF qte_fail: player_hp -= MAX(0, (enemy_atk - player_def))

   // POST-COMBAT
   IF player_hp <= 0:
      trigger_game_over()

   IF enemy_hp <= 0:
      player_exp += enemy_exp
      player_wallet += enemy_gold
      inventory.add(enemy_drop)
      end_combat()
```

Audio & SFX Systems:
Background Music
```
ON_ENTER(Map_X):
   play_bgm(Map_X_Theme)
ON_EXIT(Map_X):
   stop_bgm(Map_X_Theme)
```

Environmental SFX & Interactions
```
ON_ENTER(Map_X):
   play_ambience(Map_X_Ambience)
ON_INTERACTION(Object_X):
   play_sfx(Object_X_Sound)
```

3.3. State Machines

Player States:

- **IDLE / WALK:** Default state. Speed: 3 tiles/s. Transitions to RUN (on button press) or INTERACT.
- **RUN:** Active state. Speed: 5 tiles/s. Transitions to WALK (on button release).
- **COMBAT:** Disables world movement; executes `CombatEngine`. Transitions to DEFEATED (HP <= 0) or IDLE/WALK (Victory).

Enemy States:

- **PATROL:** Moves between Waypoints A and B.
- **PURSUIT:** Triggered if player distance < 12 tiles.
- **COMBAT:** Active during encounter.
- **DEFEATED:** Triggers "Death Animation" and removes instance from memory.

3.4. Technical Requirements (PC Target)

OS: Windows® 11 (64-bit)
Performance: 60 FPS Stable
CPU: AMD Ryzen 3 3200X (or equivalent)
RAM: 8 GB
GPU: NVIDIA GeForce GTX 1050 / AMD Radeon RX 560
Storage: 2 GB
Target Load Times: < 10 seconds